# Atmospheric Lapse Rate Regimes and Their Parameterization

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(Manuscript received 30 August 1978, in final form 22 November 1978)

ABSTRACT

Lapse rates, moist adiabatic lapse rates and the critical lapse rate for baroclinic adjustment are calculated and compared for the mean annual, January and July states in the Northern Hemisphere. In the troposphere above the planetary boundary layer zonal mean lapse rates are within 20% of the moist adiabatic lapse rate from the equator up to about 30°N in January and 50°N in July, but are appreciably more stable in higher latitudes. The latitudinal distribution of *tropospheric* mean lapse rates clearly delineates two regimes in the atmosphere—a low-latitude regime where the lapse rates are essentially moist adiabatic, and a high-latitude regime where the lapse rates are essentially the critical lapse rate for baroclinic adjustment. The dividing point between the two regimes shifts from 28°N in January to 47°N in July, and the transition is less sharp in July than in January. The absence of appreciable seasonal changes in lapse rates in midlatitudes can be attributed to counterbalancing seasonal changes in the strength of moist convection and baroclinic eddies. *Hemispheric* mean lapse rates in the mid and lower troposphere are within 0.4 K km$^{-1}$ of the moist adiabatic lapse rate in July, but are as much as 1.9 K km$^{-1}$ less in January. Implications for simple climate models are discussed. A principal conclusion is that the vertical temperature structure could be well approximated by a radiative-convective equilibrium model with *two* critical lapse rates—the moist adiabatic lapse rate and the critical lapse rate for baroclinic adjustment.

## 1. Introduction

One of the most important properties of the atmosphere that one would like to be able to predict from first principles is its vertical temperature structure. Early observations of the vertical structure indicated that the lapse rate was close to 6.5 K km$^{-1}$ with rather little seasonal or latitudinal variation (Brunt, 1933; Goody, 1964). These observations were the implicit empirical basis for a simple parameterization of the effect of dynamical fluxes on the global mean vertical temperature structure introduced by Manabe and Strickler (1964). In this parameterization the dynamical effects are included by means of a convective adjustment with the critical lapse rate taken to be 6.5 K km$^{-1}$. This empirical parameterization is frequently employed in global energy balance models for the vertical structure of the atmosphere (e.g., Pollack *et al.*, 1976; Ramanathan, 1976; Hansen *et al.*, 1978).

In fact more modern observational analyses show that the lapse rate does vary significantly, depending on the height, latitude and season (e.g., see Rennick, 1977). These variations cast doubt upon the use of a constant critical lapse rate in climate models. As a result, Rennick (1977) derived an expression in which the ratio of the static stability to the static stability of a moist adiabat at the surface was related empirically to the surface temperature. This empirical parameterization yields zonal mean lapse rates which fit all the observations quite well for surface temperatures between 250–300 K. According to the parameterization the lapse rate is nearly moist adiabatic when surface temperatures are near 300 K, but strongly sub moist adiabatic when they are near 250 K.

If one is to use such an empirical parameterization in a zonally averaged climate model, it must be physically well motivated. Rennick's parameterization is open to question on this score. As discussed by her, the most important dynamical processes that affect the lapse rate are likely to be moist convection and large-scale baroclinic eddies. The former tends to produce moist adiabatic lapse rates and the latter tend to produce more stable lapse rates. In the data the large-scale baroclinic eddies are strongest in winter in mid and high latitudes (Oort and Rasmusson, 1971). Consequently, the eddies and the more stable lapse rates are associated with low surface temperatures. However, these eddies are not driven by surface temperatures *per se*. Rather they are driven by horizontal temperature gradients and topography, which, in combination, give a maximum forcing in winter in mid and high latitudes, regardless of the surface temperatures. Consequently, the correlation between lapse rates and surface temperature discussed by Rennick is likely to be purely statistical in nature. For example, one would not expect sub moist adiabatic lapse rates to disappear in midlatitudes in January if surface temperatures rose

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to 300 K. In addition, parameterizations such as Rennick's cannot be used in *global* energy balance models, since the relation between lapse rates and surface temperatures is highly nonlinear, i.e., the zonal mean lapse rate at the latitude where the zonal mean surface temperature is equal to the global mean surface temperature is not in general the same as the global mean lapse rate.

Ideally a zonally averaged climate model would include the effect of both moist convection and large-scale baroclinic eddies on lapse rates. Recently Stone (1978) has shown empirically that the large-scale baroclinic eddies adjust the atmospheric temperature structure so that the baroclinic stability relation for a two-level model,

$$-\frac{1}{H_2}\frac{\partial T_2}{\partial \phi} \leqslant \frac{\theta_1-\theta_3}{z_1-z_3}\cot\phi, \tag{1}$$

is satisfied. Here $H$ is the scale height, $T$ the temperature, $\phi$ the latitude, $\theta$ the potential temperature and $z$ the height. The subscripts 1 and 3 refer to mass weighted values for the upper and lower halves of the troposphere, respectively, and the subscript 2 refers to the mid troposphere level. Held (1978) has supplied a theoretical rationale for this behavior. Thus if the meridional temperature structure is specified, or calculated independently, one can include the effect of the eddies on the vertical structure in the troposphere by calculating the vertical structure as though the eddies were not present, but then adjusting the structure so that Eq. (1) is not violated. The lower bound implied by (1) for the static stability in a two-layer model can be translated into a critical value of the lapse rate which is an upper bound on the tropospheric mean lapse rates which will occur. Thus the baroclinic adjustment mechanism can be simply included to a first approximation in any calculation for the zonal mean vertical structure of the troposphere by applying a convective adjustment to the tropospheric mean lapse rate, but with the critical lapse rate $\gamma_c$ taken to be the critical lapse rate implied by Eq. (1), i.e.,

$$\gamma_c \approx \gamma_d + \frac{\tan\phi}{H_2}\frac{\partial T_2}{\partial \phi}. \tag{2}$$

Here $\gamma_d$ is the dry adiabatic lapse rate.

Since moist convection is a very efficient transport mechanism, one would also expect that its effect on the vertical structure could be included to a first approximation by a convective adjustment calculation, but with the critical lapse rate taken to be the moist adiabatic lapse rate $\gamma_m$. In view of Rennick's results, one would expect this adjustment alone to give a good approximation to the lapse rates in low latitudes. However, it is not clear from her calculations how well the zonal mean lapse rate is approximated by the moist adiabatic lapse rate as a function of height, latitude and season. Also it is not clear how the two critical lapse rates $\gamma_m$ and $\gamma_c$ compare with each other. In addition, since low latitudes dominate in determining the hemispheric mean lapse rate, it is possible that a simple convective adjustment with the critical lapse rate equal to $\gamma_m$ would give good results for the vertical structure in hemispheric or global mean climate models, without having to include the effect of large-scale eddies. In order to examine these questions we will use Oort and Rasmusson's (1971) data for the zonal mean temperature structure to calculate and compare the actual zonal mean lapse rates $\gamma$ with the two critical lapse rates. The results will then be used to evaluate simple ways of parameterizing the effect of dynamics on zonal mean vertical temperature structure.

## 2. Computation of $\gamma$ and $\gamma_m$

Oort and Rasmusson's (1971) data consist of zonally averaged monthly mean values of temperature and geopotential height calculated from 5 years of observations at 11 standard pressure levels (1000, 950, 900, 850, 700, 500, 400, 300, 200, 100 and 50 mb) and at 5° latitude intervals. Thus, $\gamma$ and $\gamma_m$ could be computed on a monthly and annual basis as a function of pressure and latitude. The actual lapse rates $\gamma$ were computed simply by taking the difference between temperatures at adjacent pressure levels and dividing by the corresponding difference in geopotential height. $\gamma_m$ was calculated from the standard formula (Hess, 1959)

$$\gamma_m = \gamma_d \frac{1+\epsilon L e_s/PRT}{1+(\epsilon L/C_p P)(de_s/dT)}, \tag{3}$$

where $P$ is the pressure, $R$ the gas constant, $C_p$ the specific heat at constant pressure,

$$\frac{de_s}{dT} = \frac{\epsilon L e_s}{RT^2}, \tag{4}$$

$$e_s = e_{s0}\exp\left[\frac{\epsilon L}{R}\left(\frac{1}{T_0}-\frac{1}{T}\right)\right]. \tag{5}$$

The values used for the constants were $\gamma_d = 9.8$ K km$^{-1}$, $\epsilon = 0.622$, $R = 0.287$ J g$^{-1}$ K$^{-1}$, $C_p = 1.005$ J g$^{-1}$ K$^{-1}$, $e_{s0} = 6.11$ mb, and $T_0 = 273$ K. $L$ was computed from the expression for a perfect gas, linearized about $T = T_0$, i.e.,

$$L = 2510 - 2.38(T - T_0) \quad [\text{J g}^{-1}]; \tag{6}$$

$\gamma_m$ was computed at all pressure levels, and values at adjacent levels were then averaged for comparison with the actual lapse rate between those adjacent levels.

## 3. Latitude-height cross sections of lapse rates

Fig. 1 shows the zonal mean $\gamma_m$ as a function of pressure and latitude on an annual basis. Near the

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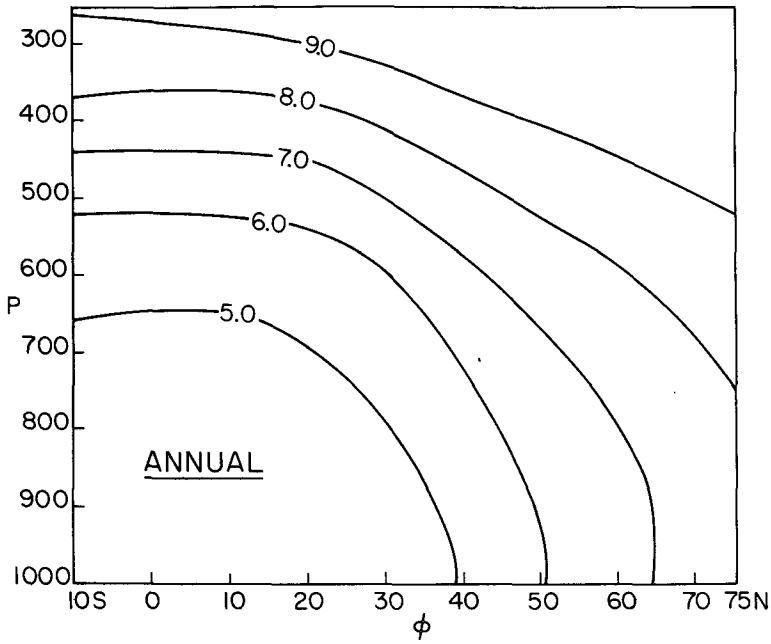



FIG. 1. Annual mean isopleths of $\gamma_m$. Units: K km$^{-1}$.

surface, values of $\gamma_m$ range from approximately 4 K km$^{-1}$ in low latitudes to 9 K km$^{-1}$ in polar regions. The $\gamma_m$ field reflects the general poleward and upward decrease of temperature. Seasonal changes in the $\gamma_m$ isopleths are small and therefore are not illustrated. These small seasonal changes reflect the seasonal changes in temperature.

Fig. 2 shows $\gamma$ as a function of pressure and latitude on an annual basis. Expected features are 1) the negative lapse rate or inversion at low levels in high latitudes; 2) the decrease of $\gamma$ with height in stratospheric polar regions; and 3) the decrease of $\gamma$ with height in the lower levels of the equatorial regions. Another noteworthy feature is the broad region between 800 and 400 mb over most latitudes for which $\gamma$ is between 5 and 7 K km$^{-1}$.

The observed lapse rates in January and July are displayed in Figs. 3 and 4. They show little departure

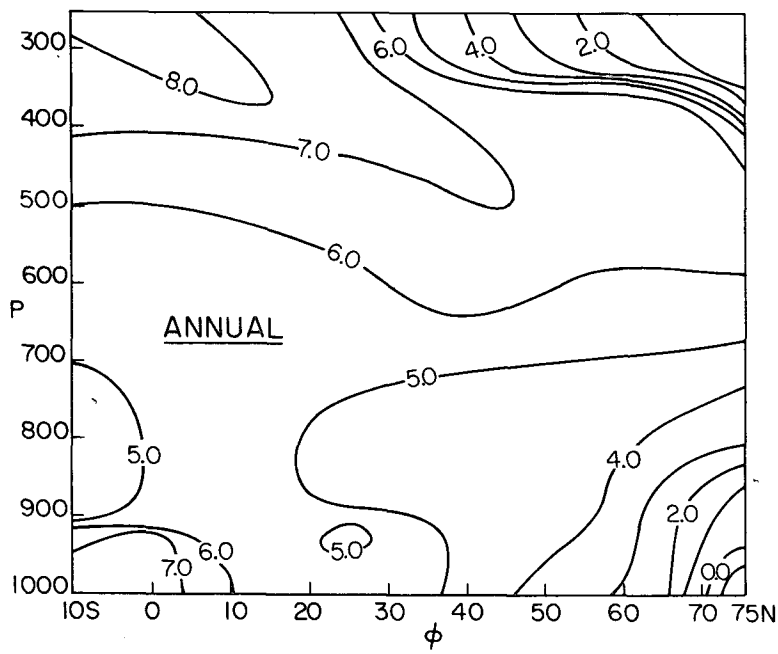



FIG. 2. Annual mean isopleths of $\gamma$. Units: K km$^{-1}$.

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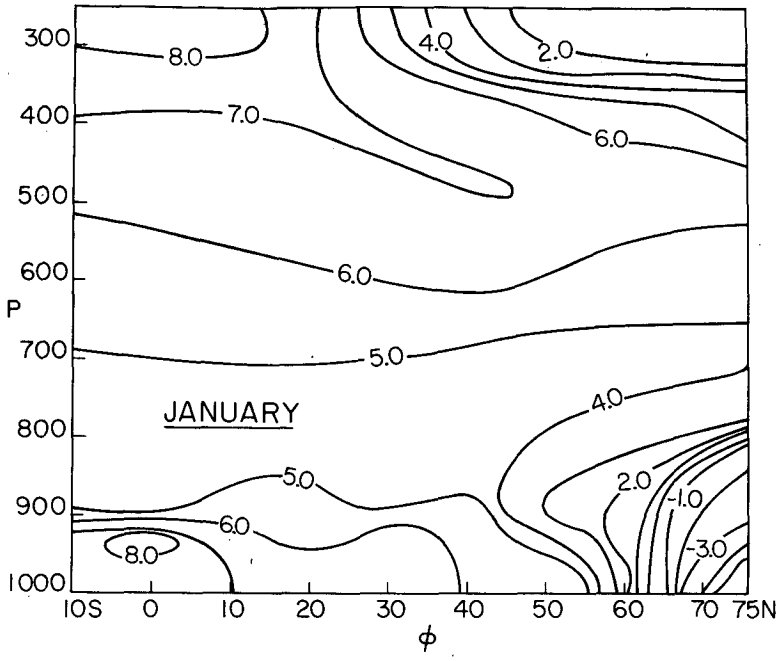



FIG. 3. As in Fig. 2 except for January.

from the annual lapse rate in the region above 700 mb and south of 30°N. North of 30°N and above 700 mb $\gamma$ is slightly larger in July than in January. Larger seasonal changes are found near the surface. For example, in the polar regions the strong inversion near the surface in January disappears in July. Large changes in $\gamma$ are also found at midlatitudes near the surface. At 40°N in January, $\gamma$ is approximately 6.0 K km$^{-1}$. This value decreases and reaches its minimum value of 2.5 K km$^{-1}$ in July. Another noteworthy feature is that south of approximately 60°N and below 850 mb, $\gamma$ is somewhat greater in January than July.

Fig. 5 gives the percentage difference between $\gamma$ and $\gamma_m$ for the annual case. One feature of interest is the poleward bulge of the isopleths at approximately 500 mb between 30° and 50°N. An expected feature is the strong conditional instability, $\gamma > \gamma_m$, below 900 mb and south of 20°N. For example, at 5°S between

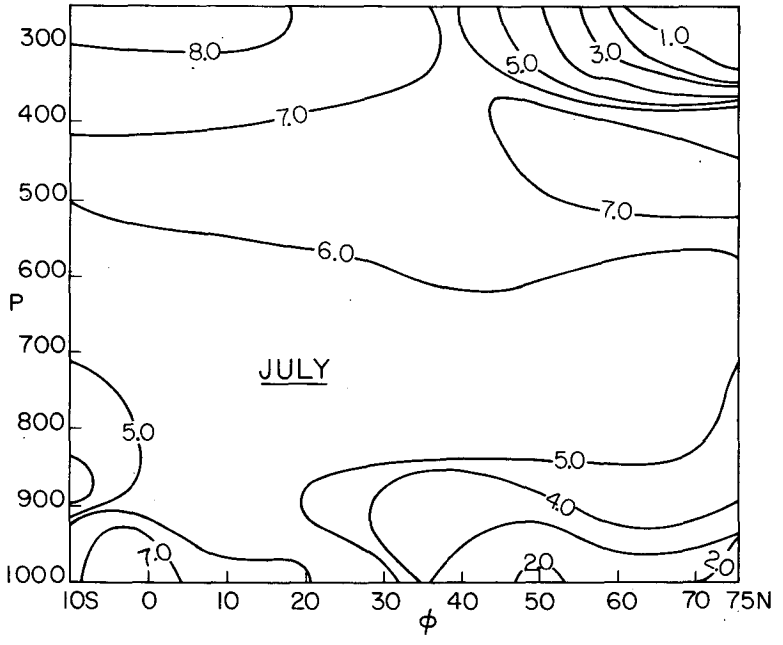



FIG. 4. As in Fig. 2 except for July.

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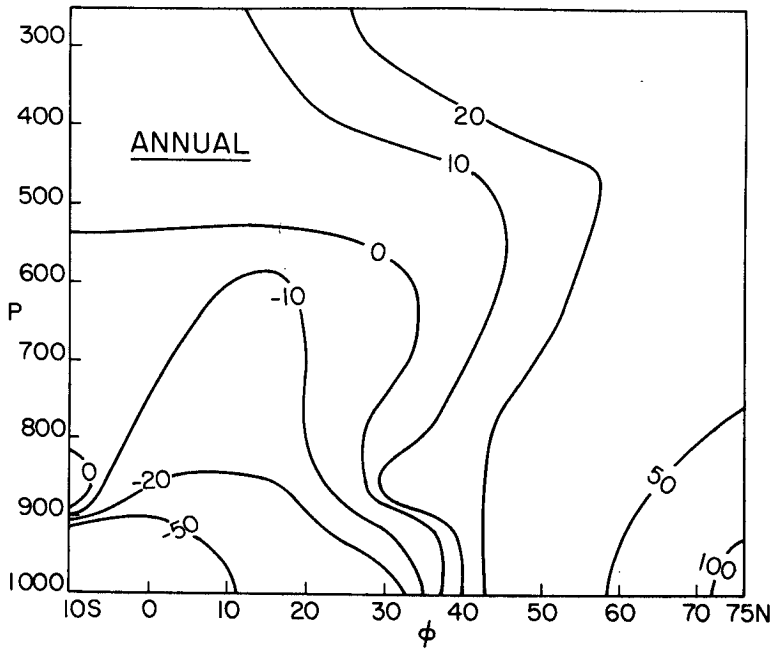



FIG. 5. Annual mean isopleths of $(\gamma_m-\gamma)/\gamma_m$ expressed as a percentage.

1000 and 950 mb, $\gamma_m=3.9$ K km$^{-1}$ and $\gamma=7.5$ K km$^{-1}$ (approximately a $-90\%$ difference). In fact, in this region $\gamma$ is closer to $\gamma_d$ than to $\gamma_m$.

Figs. 6 and 7 show the percentage differences between $\gamma$ and $\gamma_m$ in January and July. Points of interest include 1) July—the $+20\%$ isopleth extends past 75°N; 2) January and July—large differences in low levels south of 30°N; and 3) July—a stable region between 10°S and 0° at 700–900 mb, $\gamma < \gamma_m$.

The most important result illustrated by Figs. 5–7 is that the observed lapse rates are within 20% of the moist adiabatic lapse rate in low latitudes above the planetary boundary layer and below the tropopause. The northern boundary of this low-latitude region shifts from about 30°N in January to about 50°N in July. Within this region one could plausibly take the moist adiabatic lapse rate to be a first approximation to the observed lapse rates. At higher latitudes the lapse rates

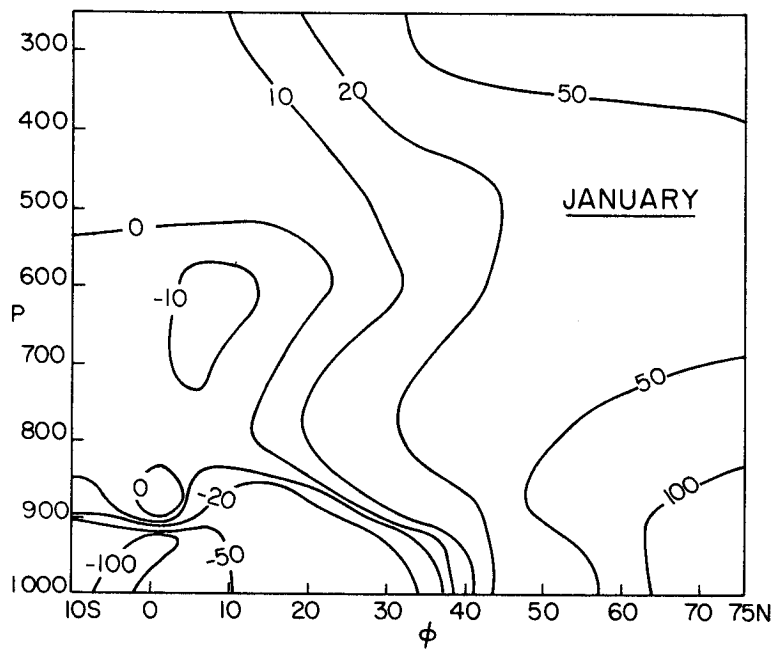



FIG. 6. As in Fig. 5 except for January.

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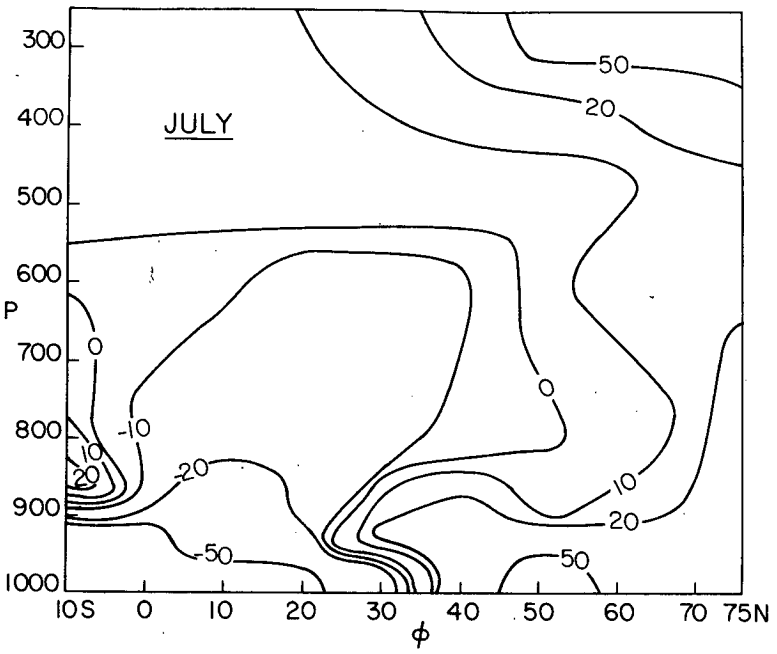



FIG. 7. As in Fig. 5 except for July.

are sub moist adiabatic and at low latitudes in the planetary boundary layer they are super moist adiabatic.

## 4. Tropospheric mean lapse rates

One particularly illuminating way of analyzing the lapse rate is to calculate tropospheric mean lapse rates and compare them with the critical lapse rates associated with moist convection and baroclinic adjustment. Mass-weighted mean values of $\gamma$ and $\gamma_m$ for the levels from 200 to 1000 mb, which are a convenient measure of the mean tropospheric values, were calculated from the data described in the preceding section. Fig. 8 shows these mean values as functions of latitude for the atmosphere's annual mean state. Also included in Fig. 8 is the critical lapse rate associated with baroclinic adjustment $\gamma_c$. In calculating $\gamma_c$ from Eq. (2) the 600 mb level was identified with level 2, and the meridional temperature gradient was calculated from finite differences across 20° latitude belts.

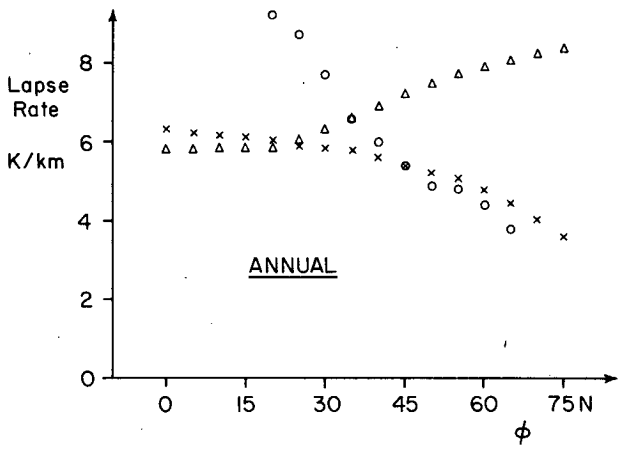



FIG. 8. Tropospheric mean values of lapse rate (×) and moist adiabatic lapse rate (△) and the critical lapse rate for baroclinic adjustment (○) as a function of latitude for annual mean conditions.

Fig. 8 illustrates how the effect on temperature structure of moist convection and baroclinic eddies varies with latitude. In low latitudes where moist convection dominates, the lapse rate agrees closely with the moist adiabatic lapse rate; and in high latitudes where baroclinic eddies dominate, the lapse rate agrees closely with the critical lapse rate for baroclinic adjustment. If we take the crossing of the $\gamma_m$ and $\gamma_c$ curves as the dividing point between the two regimes, then the dividing point is at 35°N in the annual mean state. Centered on this latitude is a small transition region, about 10 degrees wide, where both mechanisms appear to have an important effect on the temperature structure.

Figs. 9 and 10 show $\gamma_c$ and the mass-weighted values of $\gamma$ and $\gamma_m$ for the atmosphere's mean state in January and July, respectively. In both months the picture is similar to that for the annual case. The main difference is that in January the transition between the two regimes is sharper, with the dividing point being at 28°N; while in July the transition is less sharp and the dividing point is at 47°N. The transition regions are the only regions of the atmosphere where the mean lapse rate appears to be significantly less than *both* of the critical lapse rates. This depression of the lapse rate can be attributed to the combined effect of the two

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mechanisms on the temperature structure: each mechanism tends to make the atmosphere more stable in these regions than it would be if only the other mechanism were operating.

A comparison of Figs. 9 and 10 shows that the seasonal changes in lapse rate can also be simply explained in terms of the same two mechanisms. From 0°–25°N the lapse rates are controlled by moist convection in all seasons, but there is very little seasonal change in temperature. Since the moist adiabatic lapse rate is only dependent on the temperature, the lapse rates also show very little seasonal change. From 55°–75°N (and perhaps in higher latitudes) the temperature structure is controlled by baroclinic eddies in all seasons. Consequently, the lapse rates and meridional temperature gradients are strongly coupled by the baroclinic adjustment process [cf. Eq. (2)]. Thus the strong seasonal change in differential solar heating leads to substantial seasonal changes in both the meridional temperature gradients and the lapse rates from 55°–75°N. Finally in the region from 30°–50°N both mechanisms play an important role in determining the temperature structure in at least one season. Moist convection has a greater stabilizing effect in summer when the temperatures are higher but baroclinic eddies have a smaller stabilizing effect in summer because of the decreased meridional temperature gradients. Thus, the seasonal changes in the two mechanisms tend to counteract each other, and the net result is that from 30–50°N there is very little seasonal change in mean lapse rates.

## 5. Hemispheric mean lapse rates

A comparison of the hemispheric mean values of lapse rate and moist adiabatic lapse rate is particularly valuable for guidance in developing global or hemispheric energy balance models. Values of $\gamma$ and $\gamma_m$ were calculated in the manner described in Section 2 from the hemispheric mean temperature structures given by Oort and Rasmusson (1971). The resulting

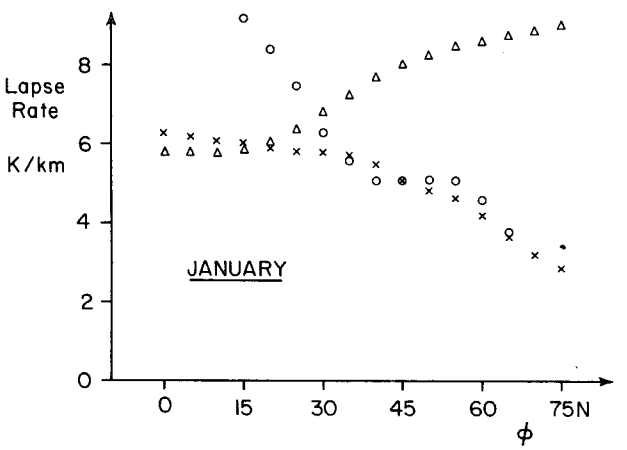



FIG. 9. As in Fig. 8 except for January.

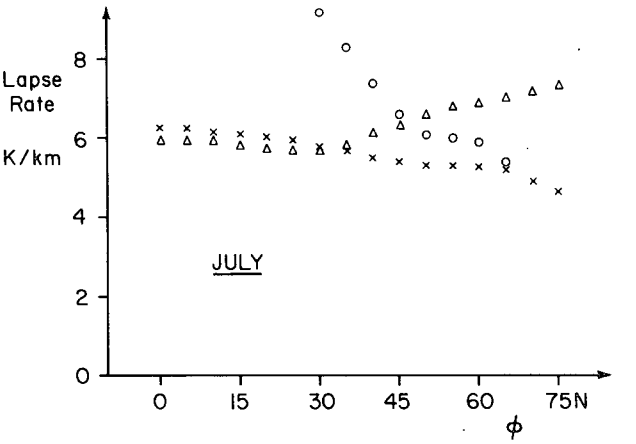



FIG. 10. As in Fig. 8 except for July.

mean lapse rates are shown in Fig. 11 as functions of pressure level for both January and July. Also shown for comparison are lapse rates of 6.5 and 9.8 K km$^{-1}$.

Fig. 11b shows that in July the moist adiabatic lapse rate is an excellent approximation to the hemispheric mean lapse rate in the middle and lower troposphere. The differences are no more than 0.4 K km$^{-1}$. This good agreement is consistent with our results in the preceding section, showing that the added stabilizing effect due to baroclinic eddies is confined to very high latitudes. Only in the upper troposphere, where the radiative destabilization is relatively weak, are the lapse rates significantly less than the moist adiabatic lapse rate.

On the other hand Fig. 11a shows that in January the hemispheric mean lapse rates are significantly less than the moist adiabatic lapse rates throughout the troposphere. In the lower troposphere they are as much as 1.9 K km$^{-1}$ less. In this season the larger region in which baroclinic eddies dominate in winter makes a significant contribution to the hemispheric mean, so that the hemispheric mean lapse rates are more stable than the moist adiabatic lapse rates even in the lower troposphere.

Fig. 11 also shows that the moist adiabatic lapse rate is a better approximation to the lapse rate in the middle and lower troposphere than the conventional approximation of 6.5 K km$^{-1}$. This result is illustrated best by a comparison of the mass-weighted mean lapse rates between 500 and 1000 mb. These mean lapse rates, calculated from the hemispheric mean temperature structures, are given in Table 1.

Even in January $\gamma_m$ is a marginally better approximation than 6.5 K km$^{-1}$ to $\gamma$. The good agreement between the annual values of $\gamma$ and $\gamma_m$ illustrates that the hemispheric mean temperature structure in the lower troposphere is dominated by moist convection. On the other hand Table 1 also shows that the seasonal change in the moist adiabatic lapse rate is much larger than the rather small change in the observed mean lapse rate and is of opposite sign. We conclude that the

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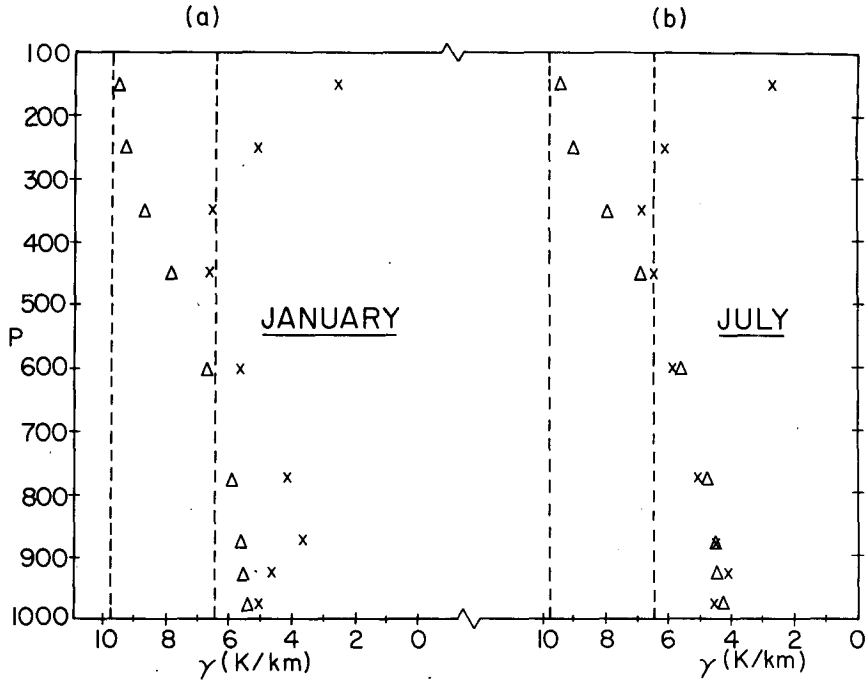



FIG. 11. Hemispheric mean lapse rate (×) and moist adiabatic lapse rate (Δ) as a function of pressure. The dotted lines indicate lapse rates of 6.5 and 9.8 K km$^{-1}$. (a) January and (b) July.

effect of seasonal changes in baroclinic eddies on seasonal changes in hemispheric mean lapse rates effectively counterbalances the effect of seasonal changes in moist convection.

## 6. Discussion

Our results for hemispheric mean lapse rates indicate that radiative-convective equilibrium models for the hemispheric mean temperature structure would give better agreement with the observations if the critical lapse rate were taken to be the moist adiabatic lapse rate rather than 6.5 K km$^{-1}$. However a model using the moist adiabatic lapse rate would give an incorrect seasonal change in the lapse rate. An examination of Fig. 11 and Table 1 might suggest that a better choice for the critical lapse rate would be a constant equal to 5.1 K km$^{-1}$. This would give accurate lapse rates in the middle and lower troposphere in both seasons. However, it is apparent from our discussion of the seasonal changes in lapse rates at different latitudes that such a choice would be unreliable for calculating climate changes. The small seasonal changes in hemispheric mean lapse rates can be attributed to counteracting seasonal changes in the strength of moist convection and of baroclinic eddies. One cannot assume that such a coincidence will occur in other kinds of climate change. In fact, our results show that even the seasonal changes in tropospheric mean lapse rate are strong north of about 50°N.

TABLE 1. Mean lapse rates in the lower troposphere (K km$^{-1}$).

| | Annual | January | July |
|---|---|---|---|
| $\gamma$ | 5.16 | 4.95 | 5.21 |
| $\gamma_m$ | 5.67 | 6.21 | 5.10 |

In order to predict changes in lapse rate a model must include the effect of both moist convection and baroclinic eddies. Since the effect of eddies on lapse rates is strongly coupled to the meridional temperature structure, one cannot avoid using at least a two-dimensional model to predict changes in hemispheric mean lapse rates. Because of this same strong coupling, predictions of the Budyko (1969) and Sellers (1969) kind of model, which treats changes in meridional temperature structure while ignoring changes in vertical structure, are also open to question.

Our comparison of lapse rates with the critical lapse rates for moist convection and baroclinic adjustment suggests a simple way of constructing a two-dimensional latitude-height model which would include both mechanisms. The vertical temperature structure at all latitudes could be found to a good approximation by using a local radiative-convective equilibrium model with *two* critical lapse rates—i.e., the lapse rate would be constrained to be less than the moist adiabatic lapse rate at all levels, and at the same time it would be constrained so that its tropospheric mean value would be less than the critical value for baroclinic

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adjustment. The mean temperature at each latitude and the meridional temperature gradients needed to calculate the critical lapse rate for baroclinic adjustment would be determined simultaneously by coupling the mean temperature at different latitudes by means of a Budyko–Sellers type of model. Such a model would allow one to study the interaction between vertical and meridional temperature structure in climate problems with a minimum of complexity.

*Acknowledgments*. This research was supported by the National Aeronautics and Space Administration under Grant NGR 22-009-727.

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